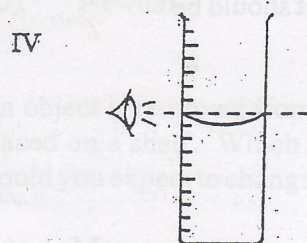
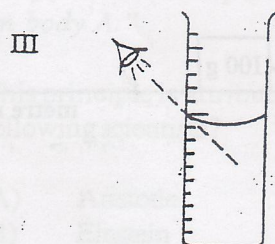
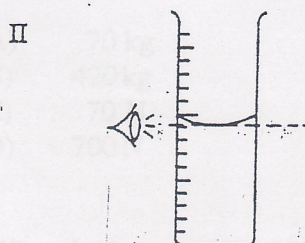
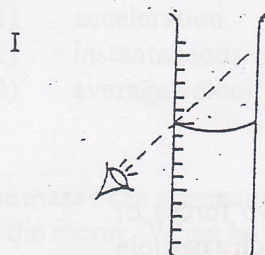


1. Which of the diagrams below indicates the correct way of taking the reading of the volume of the liquid in a measuring cylinder?



- (A) I
(B) II
(C) III
(D) IV

2. The unit of electrical resistance may be expressed as

- (A) $1\Omega = 1 \text{ V} \cdot \text{A}^{-1}$
(B) $1\Omega = 1 \text{ A} \cdot \text{V}$
(C) $1\Omega = 1 \text{ A} \cdot \text{V}^{-1}$
(D) $1\Omega = 1 \text{ W} \cdot \text{A}$

3. A student sets up a simple pendulum and finds that the period is 1.7 s. To obtain a period nearer to one second, he should

- (A) use a heavier bob
(B) use a lighter bob
(C) lengthen the pendulum
(D) shorten the pendulum

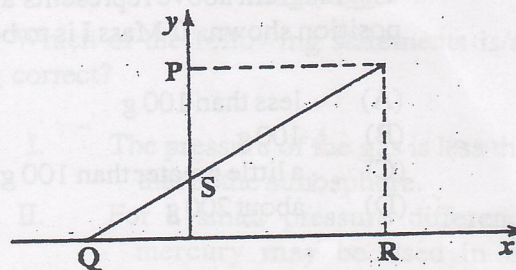
4. The instrument MOST suitable for measuring the external diameter of a test tube is a

- (A) micrometer screw gauge
(B) vernier calipers
(C) metre rule
(D) set square

5. 1 gram is equal to

- (A) 10 milligrams
(B) 100 milligrams
(C) 1 000 milligrams
(D) 10 000 milligrams

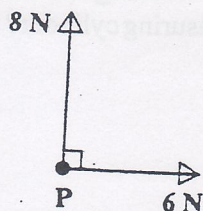
- 6.



When $x = 0$, the value of y is

- (A) Q
(B) S
(C) P
(D) P/R

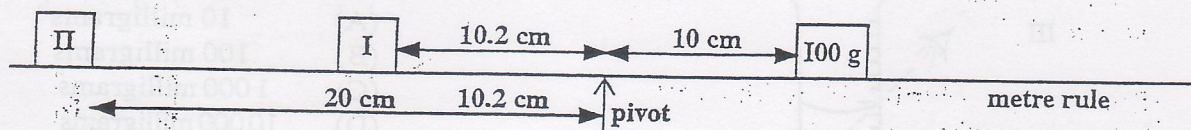
7.



The diagram above shows two forces of magnitudes 6 N and 8 N acting on a particle P. The resultant force acting on P is of magnitude

- (A) 2 N
- (B) 10 N
- (C) 14 N
- (D) 100 N

8.



The diagram above represents a 100 g mass which can be balanced by placing mass at I in the position shown. If Mass I is to be used to balance the 100 g, it should be

- (A) less than 100 g
- (B) 100 g
- (C) a little greater than 100 g
- (D) about 200 g

9.

Which of the following is NOT a fundamental quantity?

- (A) Mass
- (B) Length
- (C) Area
- (D) Time

10.

Which of the following is NOT a vector quantity?

- (A) Displacement
- (B) Acceleration
- (C) Density
- (D) Momentum

11. The slope of a velocity-time graph is a measure of

(A) distance
(B) acceleration
(C) instantaneous velocity
(D) average velocity

12. The mass of an astronaut is 70 kg when he is on the moon. When he returns to earth his approximate weight will be

(A) 70 kg
(B) 420 kg
(C) 70 N
(D) 700 N

13. "When body A exerts a force on body B, body B exerts an equal and opposite force on body A."

This principle is attributable to which of the following scientists?

(A) Aristotle
(B) Einstein
(C) Galileo
(D) Newton

14. An object is removed from the ground and placed on a shelf. Which of its properties would you expect to change?

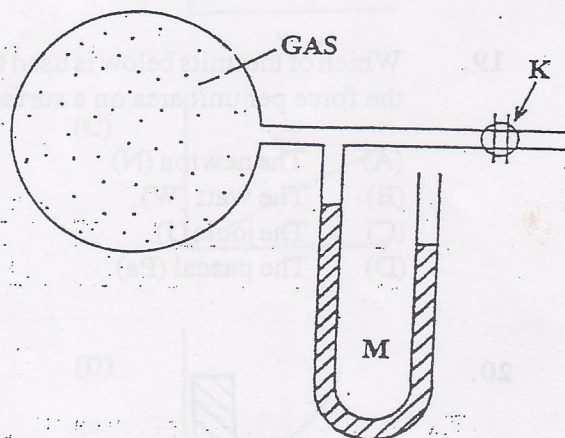
(A) Mass
(B) Volume
(C) Potential energy
(D) Kinetic energy

15. The rate of which of the following determines the power of a machine?

I. Doing work
II. Converting energy
III. Changing temperature

(A) I only
(B) III only
(C) I and II only
(D) I, II and III

16. The globe shown in the diagram below contains a gas whose pressure is measured with a manometer, M.



Which of the following statements is/are correct?

- I. The pressure of the gas is less than that of the atmosphere.
II. For a small pressure difference, mercury may be used in the manometer for an accurate reading of the gas pressure.
III. In order to bring the two menisci in the manometer to the same level one could open the stopcock, K.

(A) I only
(B) II only
(C) I and II only
(D) I and III only

17. The height of liquid in a vessel is h and its density ρ . If the atmospheric pressure is X and the acceleration due to gravity is g , what is the pressure on the base of the vessel?
(All quantities are in S.I. units.)

(A) $X + h\rho$
(B) $(X + h)\rho g$
(C) $X + h\rho g$
(D) $(X + h\rho)g$

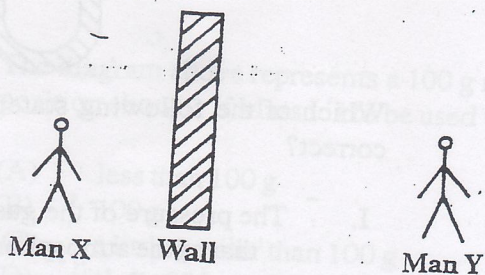
18. Which of the following would be responsible for the glass-house effect?

(A) Infra-red rays
(B) Light rays
(C) Sound waves
(D) Water waves

19. Which of the units below is used to measure the force per unit area on a surface?

(A) The newton (N)
(B) The watt (W)
(C) The joule (J)
(D) The pascal (Pa)

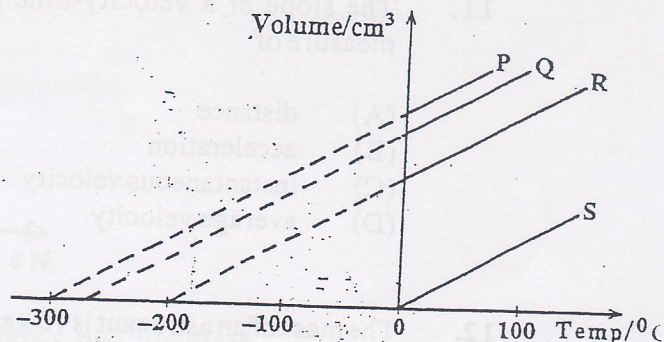
20.



Man X can hear man Y shouting but he cannot see him mainly because sound

- (A) is a longitudinal wave, and therefore does not travel in straight lines
(B) has a greater amplitude, and therefore passes around obstacles whereas light does not
(C) has a slower speed, and therefore has more time to pass around obstacles than light does
(D) has a longer wavelength, and therefore diffracts around obstacles more than light does

21.



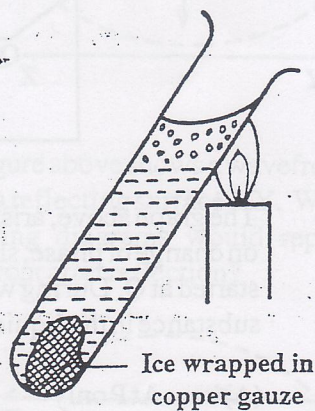
Which of the graphs above shows how the volume of a fixed mass of gas varies with temperature at constant pressure?

(A) P
(B) Q
(C) R
(D) S

22. Which of the following formulae gives the efficiency of a machine?

(A) $\frac{\text{Load} \times \text{Distance moved by Effort}}{\text{Effort} \times \text{Distance moved by Load}} \times \frac{100\%}{1}$
(B) $\frac{\text{Load} \times \text{Distance moved by Load}}{\text{Effort} \times \text{Distance moved by Effort}} \times \frac{100\%}{1}$
(C) $\frac{\text{Effort} \times \text{Distance moved by Effort}}{\text{Load} \times \text{Distance moved by Load}} \times \frac{100\%}{1}$
(D) $\frac{\text{Effort} \times \text{Distance moved by Load}}{\text{Load} \times \text{Distance moved by Effort}} \times \frac{100\%}{1}$

23.

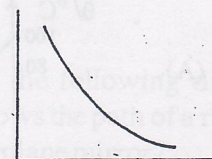


The figure above shows a piece of ice wrapped in copper gauze and submerged in a glass test tube of water. The water, when heated at the top, boils long before the ice is melted. The MAIN reason for this is that

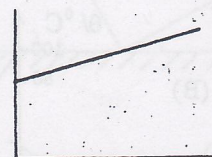
- (A) the test tube is a poor conductor of heat
- (B) copper is a good conductor of heat
- (C) ice is less dense than water
- (D) water is a poor conductor of heat

24. Which of the following graphs is obtained if the volume of a fixed mass of gas, at a constant pressure, is plotted against its Kelvin temperature?

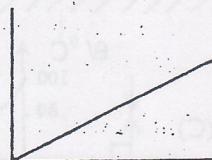
(A)



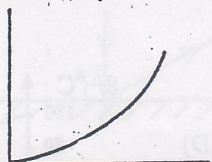
(B)



(C)



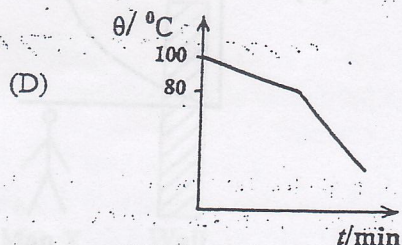
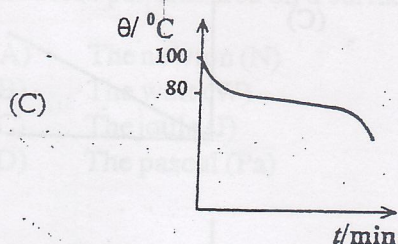
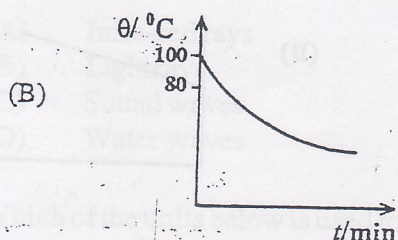
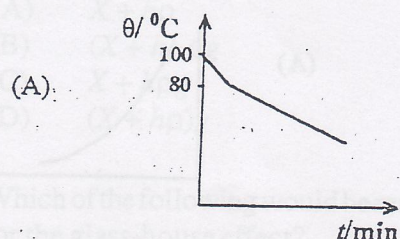
(D)



25. A flask contains air under pressure. Some of the air is let out slowly over a period of 10 s. When the flask is closed the

- (A) pressure of the air in the flask will have increased
- (B) volume of air in the flask will have decreased
- (C) temperature in the flask will have increased
- (D) number of molecules striking the wall per second will have decreased

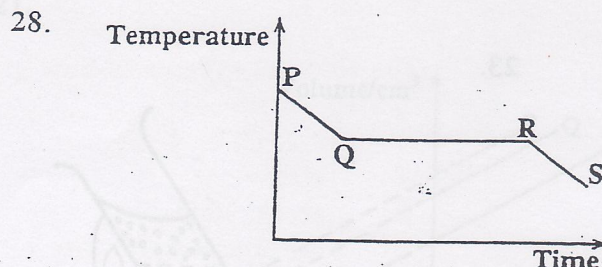
26. Some molten naphthalene at 100°C is allowed to cool down at room temperature. If naphthalene has a melting point of 80°C , which of the following BEST represents the cooling curve?



27. Which of the following waves travel only as longitudinal waves?

- I. Sound waves
- II. Radio waves
- III. Water waves

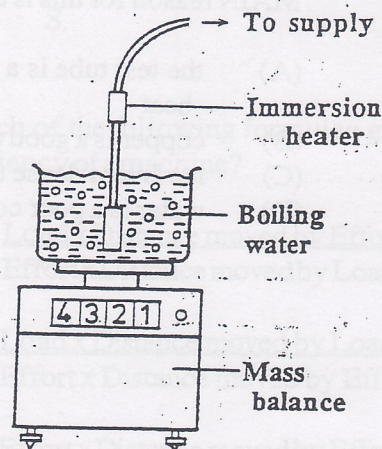
- (A) I only
- (B) II only
- (C) II and III only
- (D) I, II and III



The graph above, arising from an experiment on change of phase, shows that solidification started at Q. During which of the stages is the substance in the liquid phase?

- (A) At P only
- (B) Between Q and R
- (C) Between R and S
- (D) Between P and Q

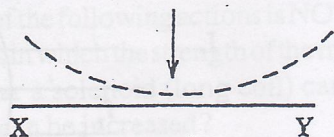
29.



Some water in a container (as shown above) is brought to its boiling point by an immersion heater. A watch is then started and the time, t , to boil off a mass, m , of water is found. If the heater has a power, P , the specific latent heat of vaporisation is

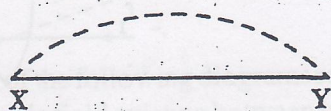
- (A) $\frac{Pt}{m}$
- (B) $\frac{P}{mt}$
- (C) $\frac{m}{Pt}$
- (D) $\frac{Pmt}{m}$

30.

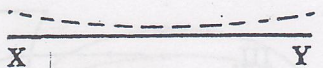


The figure above shows a wavefront about to reach a reflecting surface XY. Which of the following diagrams would represent the wavefront after reflection?

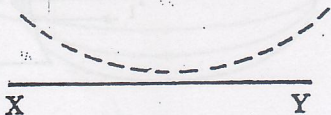
(A)



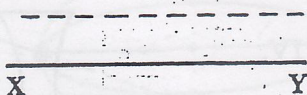
(B)



(C)



(D)



31. The note from a drum is louder when it is struck harder because the sound waves produced have a greater

- (A) amplitude
- (B) frequency
- (C) wavelength
- (D) velocity

32. Compared with X-rays, radio waves

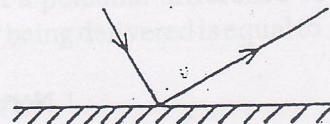
- (A) have a longer wavelength
- (B) have a higher frequency
- (C) travel faster in air
- (D) pass through a sheet of steel more easily

33. The sharp edges of shadows suggest that light

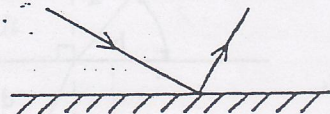
- (A) has a wave nature
- (B) is a form of energy
- (C) travels in straight lines
- (D) travels very quickly

34. Which of the following diagrams MOST clearly shows the path of a ray of light when it strikes a plane mirror?

(A)



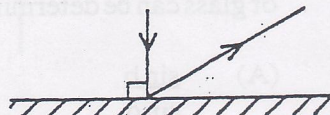
(B)



(C)



(D)



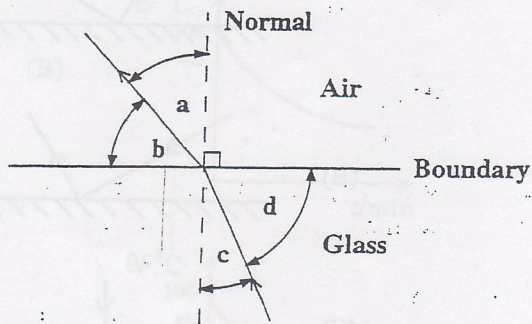
35. In a projector the object is 5 cm from the lens, while the screen is 500 cm from the lens on the other side. What is the magnification?

- (A) 50
- (B) 100
- (C) 250
- (D) 500

36. When heat is supplied at equal rates to equal masses of water and of cooking oil, the cooking oil attains a higher temperature than water in the same time. This is because

- (A) water has a higher specific latent heat
- (B) water has a higher specific heat capacity
- (C) oil is a better conductor of heat
- (D) water is a better conductor of heat

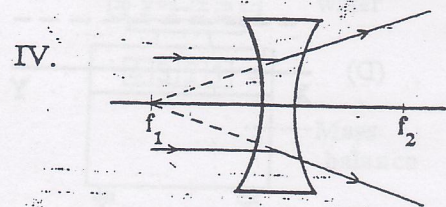
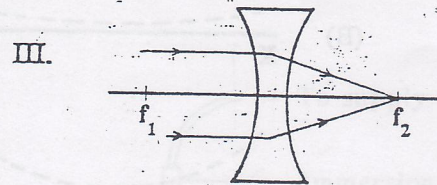
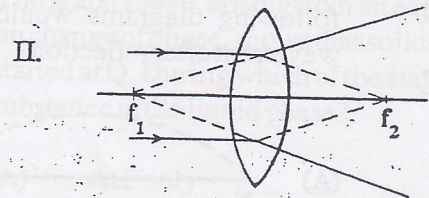
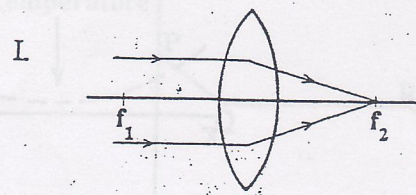
37.



The diagram above represents a ray of light passing from glass to air. The refractive index of glass can be determined by calculating

- (A) $\frac{\sin b}{\sin d}$
- (B) $\frac{\sin a}{\sin c}$
- (C) $\frac{\sin c}{\sin a}$
- (D) $\frac{\sin b}{\sin c}$

38.



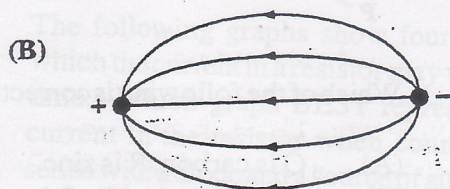
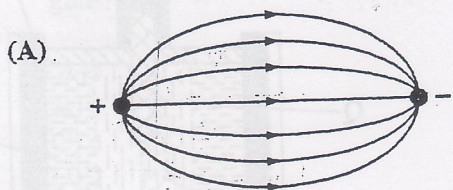
Which of the ray diagrams shown above are correct?

- (A) II and III only
- (B) II and IV only
- (C) I and III only
- (D) I and IV only

39. Which of the following actions is NOT one of the ways in which the strength of the magnetic field near a solenoid (long coil) carrying a current can be increased?

- (A) Increasing the resistance of the coil
- (B) Increasing the current in the coil
- (C) Increasing the number of turns per unit length of the coil
- (D) Placing a soft iron core inside the coil

40. Which of the following diagrams represents the electric field existing between two oppositely charged point charges?



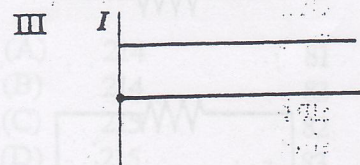
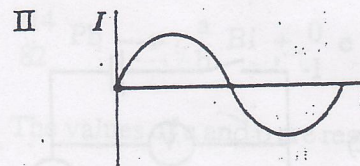
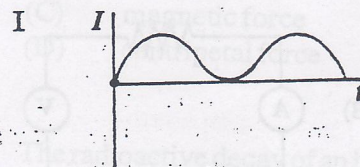
41. Which of the following equations is NOT correct?

- (A) $1 \text{ C} = 1 \text{ A} \times 1 \text{ s}$
- (B) $1 \text{ V} = 1 \text{ A} \times 1 \Omega$
- (C) $1 \text{ J} = 1 \text{ C} \times 1 \text{ V}$
- (D) $1 \text{ W} = 1 \text{ V} \times 1 \text{ C}$

42. A current I flows for a time t between two points at a potential difference of V . The power P being delivered is equal to

- (A) IV/t
- (B) IVt
- (C) IV
- (D) It

43. Which of the following current-time graphs would represent direct current?

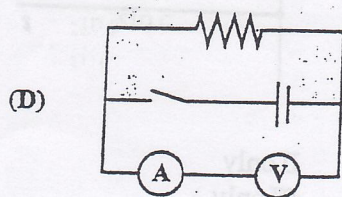
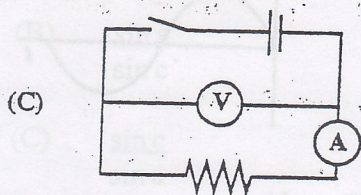
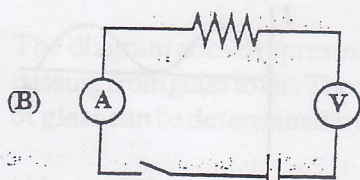
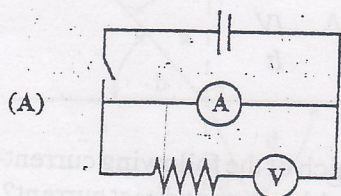


- (A) II only
- (B) III only
- (C) I and III only
- (D) I, II and III

44. A vibrator on a ripple tank sends waves over the surface of the water. The vibrator is now adjusted so that the number of waves created each second is doubled. The waves would now

(A) travel at half the original speed
 (B) travel at twice the original speed
 (C) have twice the original wavelength
 (D) have half the original wavelength

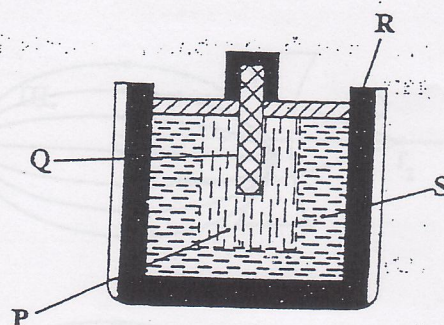
45. A student requires a circuit to measure the resistance of a resistor. Which of the circuits below is correctly connected?



46. An ammeter has a very low resistance so that it can be placed

(A) in parallel with a component and not affect the circuit
 (B) in series with a component and not affect the circuit
 (C) in parallel with a component and the ammeter does not heat up
 (D) in series with a component and the ammeter does not heat up

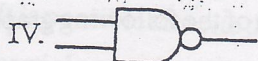
Item 47 refers to the diagram below of the cross-section of a dry cell.



47. Which of the following is correct?

(A) Q is carbon, R is zinc.
 (B) P is copper oxide, Q is carbon.
 (C) R is iron, S is ammonium chloride.
 (D) S is sulphuric acid, P is manganese dioxide.

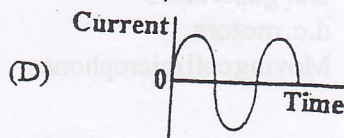
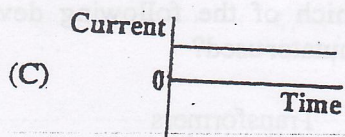
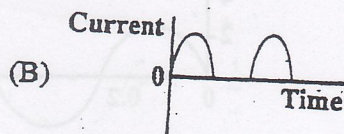
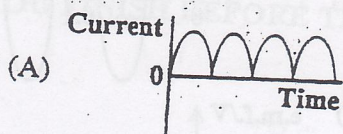
Item 48 refers to the following symbols.



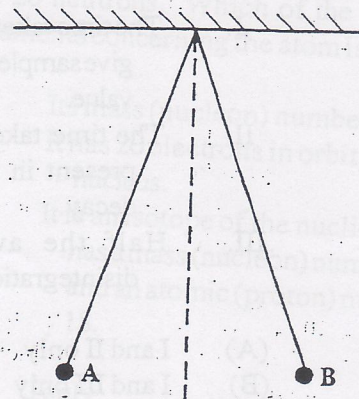
48. Which of the symbols represents the AND gate?

- (A) I
(B) II
(C) III
(D) IV

49. The following graphs show four ways in which the current in a resistor may vary with time. Which graph BEST represents the current in the resistor when connected in series with a diode and a source of alternating current?



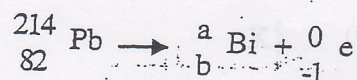
50.



Two light, aluminium spheres A and B are suspended by insulating threads. If they come to rest as shown in the diagram, the force keeping them apart is

- (A) gravitational force
(B) electrostatic force
(C) magnetic force
(D) centripetal force

51. The radioactive decay of an isotope of lead is represented by the equation



The values of a and b are respectively

- | | a | b |
|-----|-----|----|
| (A) | 214 | 81 |
| (B) | 214 | 83 |
| (C) | 215 | 82 |
| (D) | 215 | 83 |

52. Which of the following are definitions of the term 'half-life' of radioactive nuclide?

- I. The time taken for the activity of any give sample to fall to half its original value.
- II. The time taken for half the nuclei present in any given sample to decay.
- III. Half the average number of disintegrations per second.

- (A) I and II only
- (B) I and III only
- (C) II and III only
- (D) I, II and III

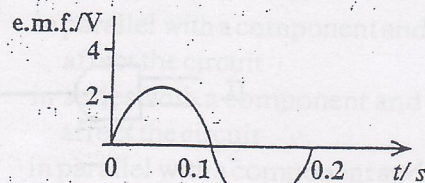
53. Which of the following is an isotope of $^{12}_6\text{X}$?

- (A) ^6_3X
- (B) $^{13}_6\text{X}$
- (C) $^{12}_{10}\text{X}$
- (D) ^3_1X

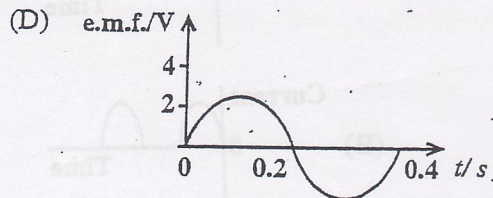
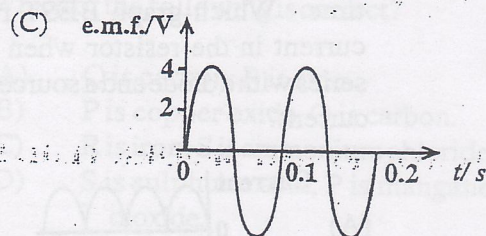
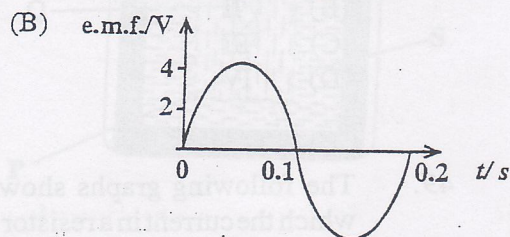
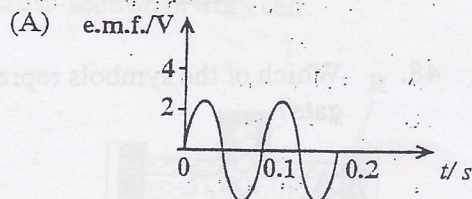
54. An isotope of zinc has a nuclide which can be represented as $^{64}_{30}\text{Zn}$. The number of electrons in a neutral zinc atom is

- (A) 30
- (B) 34
- (C) 64
- (D) 94

55. The graph below shows how the e.m.f. at the output of an a.c. generator varies with time.



If the frequency of this waveform is doubled which of the following graphs is produced?



56. In which of the following devices is a commutator used?

- (A) Transformers
- (B) a.c. generators
- (C) d.c. motors
- (D) Moving coil microphones

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57. The Rutherford model of the atom suggests that an atom consists of a

- (A) solid mass of protons and electrons
- (B) nucleus of protons only with orbiting electrons
- (C) nucleus of equal numbers of neutrons and electrons with orbiting protons
- (D) nucleus of protons and neutrons with orbiting electrons

58. It is possible to write the symbol for a certain particle x as 1_0x . This particle is a/an

- (A) electron
- (B) neutron
- (C) proton
- (D) alpha particle

59. A neutral atom of Argon contains 18 protons and 20 neutrons. Which of the following statements concerning the atom is/are true?

- I. Its mass (nucleon) number is 38.
- II. It has 20 electrons in orbit about the nucleus.
- III. It is an isotope of the nuclide which has a mass (nucleon) number of 40 and an atomic (proton) number of 18.

- (A) I only
- (B) I and II only
- (C) I and III only
- (D) I, II and III

60. Radiation coming from a radioactive source was deflected by a magnetic field and it was stopped by a few millimetres of air. The radiation was

- (A) neutrons
- (B) gamma rays
- (C) beta-particles
- (D) alpha-particles

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.